Center For Advanced Studies in Engineering

Crime Detection in Local Languages

DEPARTMENT OF COMPUTER SCIENCE

FINAL YEAR PROJECT REPORT

SUBMITTED BY

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in Partial Fulfillment of the Requirement
for the Degree of
Bachelor of Science in Computer Science

DEPARTMENT OF COMPUTER SCIENCE

Sir Syed CASE Institute of Technology

Islamabad

2021

CERTIFICATE OF APPROVAL

This project titled

Crime Detection in Local Languages

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By

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DECLARATION

We the undersigned solemnly declare that the project report CRIME DETECTION IN LOCAL LANGUAGES is based on our own work carried out during the course of our study under the supervision of Ms. Saira Qamar. We assert the statements made and conclusions drawn are an outcome of our research work.

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I certify that the work in this thesis is carried out and completed under my supervision.

Signature: _____

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ABSTRACT

Research in the domain of crime detection in low resource languages such as Urdu is a field that is not explored extensively. Mostly, due to the unavailability of Urdu data we face the challenges of generating and handling unlabeled Urdu audio data. This research serves the purpose of predicting and preventing crime based on data gathered through the surveillance of audio channels. There are speech recognition systems developed for Urdu, whereas this paper presents a model that uses an unsupervised approach to apply sentiment analysis to a speech corpus. The objective of our work is to apply natural language processing (NLP) techniques onto an unlabeled dataset. The data is then embedded on the basis of each word, which is then added together to calculate the average sentence vectors which are plotted so that the process of clustering can be applied.

The results of our research show that our model approximately identifies sentences based on our lexicon based sentiment analysis model through the proper use of clustering method, by making different clusters of our self-generated lexicons in order for our model to distinguish between sentences.

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CHAPTER 1

1. INTRODUCTION

1.1 Brief

The purpose is to create an AI solution to detect crime in Urdu audio clips, which would ultimately help our law enforcement agencies in eradicating and stopping the crime before it even happens. A conversation between two individuals can be monitored and our model would detect whether the discussion being done is related to any sort of crime or not.

1.2 Relevance to Course Modules

One of the key components of this project was data preprocessing which we learnt through the Natural Language Processing Course. We have used clustering for context checking which we learned at some level in Artificial intelligence course. As shown in Table. 1.

Concept Used	Relevance to Course
Data Clustering	Artificial Intelligence
Data Pre-Processing	Natural Language Processing
Front End Development	Web Designing

Table 1 Concept Relevance to Course Module

1.3 Project Background

Very little research has been done on using our local languages to detect crime in the country. We have a vast supply of unfiltered data that can be used to train even the most complex systems. We aim to target this area and design a model that utilizes local languages to detect crime/slang language in conversations.

1.4 Software Lifecycle

We use general software lifecycle for developing of our project as shown in Figure. 1.

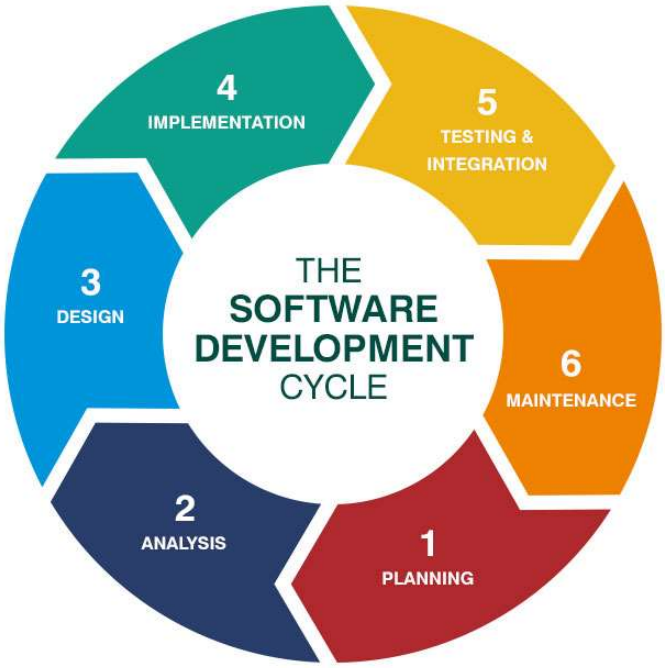

Figure 1 Lifecycle

Planning:
The initial plan involved creating an Urdu speech corpus to be used as the dataset for the model. Once an adequate amount of data was gathered, the next step was to clean the data to be able to run our detection model on it. Dealing with unsupervised data meant we had to apply clustering on the data to be able to deduce context.

Analysis:
The analysis phase involved researching previous works in the field and choosing which direction to take the project in that would help the learn as much as possible.

Design:
We designed the pre-processing module to clean the data. The data is then passed through the sentiment analysis and clustering modules to gather context and then displayed on the front end for the user.

Implementation:
We implemented all the necessary functions necessary to fulfill the requirements of the project. We used the support of API's to manage some of the heavier tasks.

Testing and Integration:
Each module was designed and implemented separately so the final step was integrating them to receive data from each other smoothly without data loss. Testing was done through clustering validation.

Life Cycle	Description
Planning	How to collect data and gather context from audio.
Analysis	We analyze which techniques for sentiment analysis should be used.
Design	Design the algorithms and front end for display.
Implementation	Implementation is done using Python.
Testing and Integration	Validate clustering.

Table 2 Lifecycle

CHAPTER 2

2. PROBLEM DEFINITION

2.1 Problem Statement

Language processing is at a very basic level when it comes to Urdu and for this very reason; this paper targets the Urdu language as its primary local language to develop a system that deduces sentence context based on just audio.

2.2 Literature Review

Sentiment Analysis is a field that has extensive work being done on it, but mostly on high resource languages. Not many resources are available for languages like Urdu. So, for the purpose of research we looked at how sentiment analysis was catered to languages other than English i.e. Spanish, Arabic etc. Our project has such specific requirements, we split our research based on modules and looked through papers that covered various parts that would ultimately be combined to form the final product. Speech corpus generation, Data Pre-processing, Sentiment Analysis, Handling unsupervised data, Clustering of data were the main modules we focused on researching.

Speech corpus is an essential component of any speech interface system. The development and cleaning of the data to ensure the model only analyses the useful parts of a sentence before giving the result.

Speech Recognition

Automatic Speech Recognition (ASR) is one of the applications of speech and language technologies that convert speech into text. Urdu is the national language of Pakistan; bridging people speaking regional languages such as Balochi, Pashto, Punjabi and Sindhi with various dialects. Urdu is a low resource language and very little transcribed speech data, text corpus and pronunciation lexicon is available publicly. A robust speech recognition system requires hundreds of hours of transcribed speech data, very large text corpus and lexicon. Speech recognition for low resource languages has received less attention in recent few years. [2] The term “under-resourced languages” introduced by [Krauwer, 2003 ; Berment, 2004] refers to a language with some of (if not all) the following aspects: lack of a unique writing system or stable orthography, limited presence on the web, lack of linguistic expertise, lack of electronic resources for speech and language processing. The synonyms for the same concept are: *low-density languages, resource-poor languages, low-data languages, less-resourced languages.*
Speech recognition systems are categorized into three types; isolated word, connected word and continuous speech recognition systems. Isolated word systems recognize only isolated word, connected word systems recognize connected words while continuous speech recognition systems recognize words spoken as in natural conversation.
Speech recognition for resource rich languages has excelled greatly but it is still at a rudimentary level for under resource languages such as Urdu. In order to develop speech recognition systems for under resource languages, Speech-based Automated Learning of Accent and Articulation Mapping (SALAAM) [3] uses the existing avail-

able resources of “developed languages” like English to develop systems for “developing languages”. The SALAAM method was tested for 10 diverse languages and yielded word recognition rate around 90% for 3 to 10 words. An improved method of speech recognition for low resource languages based on SALAAM is presented in [4]. They used US English as source language and target languages included Yoruba, Hindi and Hebrew. They reported word recognition rate up to 90% for vocabulary size of 20-30 words but it drops sharply for higher vocabulary size (word recognition rate of 50 word vocabulary for Hindi is around 77%).
The development of an isolated speech recognition system for Urdu language is described in [5]. The vocabulary size was 52 words and training data was collected from ten speakers. The word recognition rate for the speakers present in training data is 94.67%. For speakers not present in training data, word recognition rate is 89.34%. The system is not feasible for a real word application as the data used is statistically insignificant. A first attempt to develop a continuous speech recognition system for Urdu language is discussed in [6]. The corpus for the system consisted of data recorded from 42 male and 40 female speakers with total duration of 45 hours. The reported word recognition rate was around 40% due to insufficient noise modelling, diacritic issues and lack of accent variation handling.

Urdu Speech Corpus

While a lot of research has been conducted in developing annotated speech corpora of various languages, only limited work has been conducted for Urdu language speech corpus development [7]. The U.S. Army Research Laboratory (ARL) Urdu Speech Database, a collection of recorded speech from 200 adult native Urdu speakers from Pakistan and Northern India, is available through the Linguistic Data Consortium (LDC) [8]. The corpus was released in February 2007. Its data is divided into training and testing sets, and has speech of multiple Urdu dialects including South Sindh, North Sindh, South Punjab, North Punjab, North West regions and Baluchistan. Speech corpora development in Urdu and other Indian languages for speaker recognition is also reported in [9].

Data Pre-Processing

Majority of the recent research works on social media Arabic text that provide solutions to the problems encountered in social media Arabic text processing focus on preparing data to specific objectives, mainly sentiment analysis. However, some of these studies presented satisfactory solutions in the area of social media Arabic text processing, e.g., Darwish et al. [27] improved the retrieval of Arabic text from micro blogs by enhancing the normalization of Arabic micro blogs on Twitter. They first considered the normalization of Arabic letters, and then expanded their normalization processes to handle words using processes such as normalizing elongated and shortened words, removing stop words, stemming, and considering Arabic dialects. Al-Twairesh et al. [26] collected the dataset for their study from Arabic tweets and then applied existing methods to clean and preprocess the collected data set. They removed the URLs and user mentions (@user) and normalized the Arabic letters (أ,ة,ي,و). Then, they annotated the data for sentiment analysis.

Recently, deep learning techniques have been used in sentiment analysis research. Some of these studies use preprocessing or NLP in preparing their data. For example,

in [28], the text was tokenized and then translated into English before applying the deep learning algorithms. In [29], all diacritics Arabic text was removed. The stop words, punctuations, numbers, symbols, and non-Arabic words were replaced by keywords. Only three Arabic characters (ى,ة,أ) were normalized.

[30] Presents efficient preprocessing techniques. These pre-processing techniques eliminates noisy from text data, later identifies the root word for actual words and reduces the size of the text data. This improves performance of the IR system.

Lexicon based sentiment analysis

Sentiment analysis is considered as a classification problem [10]. The text is classified as positive or negative. Some classifiers use a multi-point-range e.g., five star scale for movie reviews, etc. The classification usually starts from term or phrase level and moves towards sentence and then document level. Usually, the output of one level becomes input to the next [11]. There are, a number of features on the basis of which, the text is classified. For example, frequency and position of a term or information of a part of speech like adjectives and adverbs [3]. Approaches based on lexicons of words/phrases or expressions, are pre-tagged with sentiment, subjectivity or polarity information about each entry. This lexicon can be manually created as well as automatically generated, using machine learning methods for example [10] and [13]. Some researchers have used WordNet for sentiment mining like [14]. In [15] a lexicon of around 21556 words was developed. The developed lexicon was manually tested and a 75% inter-annotator agreement was achieved. Around 25% data showed a mismatch of polarities due to the subjectivity of the annotator and the influence of the pragmatic stance during the data tagging process. In some cases, work can be done on phoneme, word, syllable, and break index levels like in [16].

Each word in the text (to be analyzed) is compared with the lexicon entry. As a result a positive or negative orientation score is attached to it. The total score of a sentence is then computed as a sum of word scores.

Clustering

Cluster analysis [17] is one of the major data analysis methods which are widely used for many practical applications in emerging areas like Bioinformatics etc. Clustering is the process of partitioning a given set of objects into disjoint clusters. This is done in such a way that objects in the same cluster are similar while objects belonging to different clusters differ considerably, with respect to their attributes. One clustering technique is the method of K-means algorithm using the process repeatedly. The K-Means method is the simplest and most common clustering method. K-means has the ability to group large amounts of data with relatively fast and efficient computation time [21]. However, K-Means has a disadvantage depending on the initial cluster center determination. K-Means cluster test results in the form of solutions that are local optimal. From the trial process is expected to have similarities or closeness between data so that can be grouped into several clusters, where among cluster members have a high level of similarity [22].

As shown in the Algorithm, the k-means algorithm consists of two phases: one for determining the initial centroids and the other for assigning data points to the nearest clusters and then recalculating the cluster means. The second phase is carried out repetitively until the clusters get stabilized, i.e., data points stop crossing over cluster boundaries. Fang Yuan et al. [18] proposed a systematic method for finding the initial centroids. The centroids obtained by this method are consistent with the distribution of data. Hence it produced clusters with better accuracy, compared to the original k-means algorithm. However, Yuan's method does not suggest any improvement to the time complexity of the k-means algorithm.

According to (Celebi et al., 2013) the K-Means algorithm is also versatile, which is easy to modify at every stage of the process, simple in the distance calculation function, and depends on iteration termination criteria. K-Means Clustering is a localized optimization method that is sensitive to the selection of the starting position from the midpoint of the cluster. So choosing the starting position from the midpoint of a bad cluster will result in K-Means Clustering algorithm being trapped in the optimal locale [23].

2.3 Deliverables and Development Requirements

Deliverable	**Development Requirements**
Dataset	Urdu audio data compiled using microphones and online sources.
Pre-Processed Data	Clean the data and shrink it down into a form that is easy to classify.
Clustering Reports	Find context in the data using clusters.
Front End	Develop a front end to display the results.

Table 3 Deliverables

2.4 Current Systems

Currently no open source system exists with this exact specification. There are systems that run sentiment analysis on Urdu, but not in the context of crime.

CHAPTER 3

3. REQUIREMENT ANALYSIS

3.1 Use Case Diagram

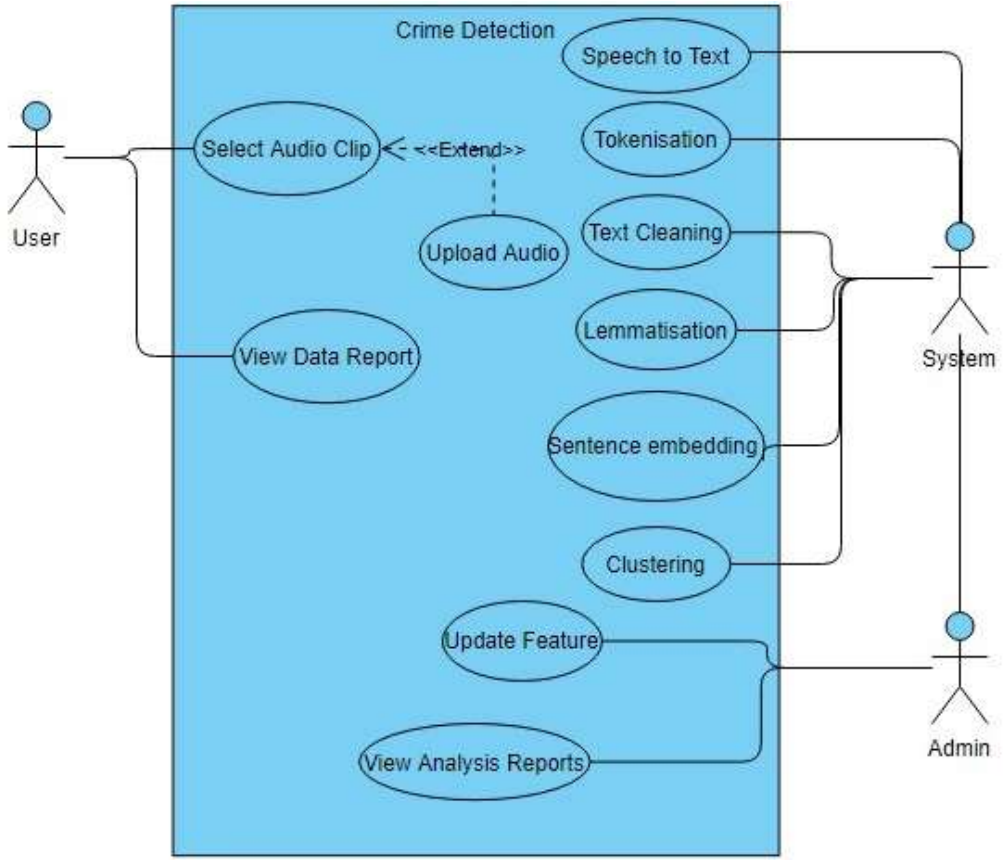

Figure 2 Use Case

3.2 Use case name and identifier (Use Case Narrative)

1.	Use Case Name: Speech to Text	
2.	Implementation Priority: 1	
3.	Actors: Admin	
4.	Summary: Admin runs Google Speech Recognition API to convert audio to text.	
5.	Pre-condition: Admin must have Urdu audio dataset.	
6.	Post-condition: The textual data will be passed to preprocess module.	
7.	Extend: -	
8.	Include: -	
9.	Normal Course of Events:	
	System Actor Action	System Response
	1. Admin inputs audio files. 4. Admin can use the text for next step.	2. System runs the files through the API, returning text. 3. Text is stored in a CSV file.
10.	Alternative Path: Using another algorithm apart from the API for conversion.	
11.	Exception: Audio format clashes with API.	

Table 4 Speech to Text Narrative

1.	Use Case Name: Tokenization	
2.	Implementation Priority: 2	
3.	Actors: Admins	
4.	Summary: Admin runs designed tokenization algorithm which breaks the sentence into tokens.	
5.	Pre-condition: Admin must have access to the text data.	
6.	Post-condition: Each word in the sentence will be considered a token.	
7.	Extend:-	
8.	Include: -	
9.	Normal Course of Events:	
	System Actor Action	System Response
	1. Admin runs the code. 4. Admin can now work on the array list.	2. System breaks each sentence into tokens based on spaces. 3. Tokens are saved in an array list.
10.	Alternative Path:-	
11.	Exception: -	

Table 4 Tokenization Narrative

1.	Use Case Name: Text Cleaning	
2.	Implementation Priority: 3	
3.	Actors: Admins	
4.	Summary: Remove white spaces, punctuation, stop words and special characters from the list.	
5.	Pre-condition: Data should be in the form of tokens.	
6.	Post-condition: Clean data is available for processing.	
7.	Extend: -	
8.	Include: -	
9.	Normal Course of Events:	
	System Actor Action	System Response
	1. Admin runs text cleaning algorithm. 5. Admin can now use the clean data that has no extra information.	2. System compares the tokens to values from predefined lists and removes those values.
10.	Alternative Path:	
11.	Exception: Lists are not complete and so not every unnecessary token is removed.	

Table 5 Text Cleaning Narrative

1.	Use Case Name: Lemmatization	
2.	Implementation Priority: 4	
3.	Actors: Admin	
4.	Summary: Break down words into their base forms.	
5.	Pre-condition: Data should be tokenized and cleaned.	
6.	Post-condition: All big words shrunk to base forms and easier to process.	
7.	Extend: -	
8.	Include: -	
9.	Normal Course of Events:	
	System Actor Action	System Response
	1. Admin runs algorithm code. 3. Admin can now work on only the important bits of the sentence.	2. System compares tokens to the lemmatizing lookup table. And converts values to base form.
10.	Alternative Path: Pick audio files that only focus on base form of words.	
11.	Exception: Lookup table is not vast enough to cater every word.	

Table 6 Lemmatization Narrative

1.	Use Case Name: Sentence Embedding	
2.	Implementation Priority: 5	
3.	Actors: Admin	
4.	Summary: Convert textual tokens into vectors to be plotted.	
5.	Pre-condition: Data should be tokenized and cleaned.	
6.	Post-condition: Each sentence will be converted into average sentence vectors which can be used for mathematical calculations.	
7.	Extend: -	
8.	Include: -	
9.	Normal Course of Events:	
	System Actor Action	System Response
	1. Admin passes sentences through word2vec model. 3. Admin can now plot the sentences.	2. Model converts words into 300 dimension vectors which are then averaged on the basis of sentences.
10.	Alternative Path: Using a term frequency- inverse document frequency (TFIDF)	
11.	Exception: The pre-trained model is not compatible with our work.	

Table 8 Sentence Embedding Narrative

1.	Use Case Name: Clustering	
2.	Implementation Priority: 6	
3.	Actors: Admin	
4.	Summary: Classify data based on their similarities to surrounding data.	
5.	Pre-condition: Data should be in vector form.	
6.	Post-condition: Graphs showing all the similar sentences clustered together.	
7.	Extend: -	
8.	Include: -	
9.	Normal Course of Events:	
	System Actor Action	System Response
	1. Admin runs clustering algorithm. 4. Admin can now analyze the data based on graphs and reports.	2. System applies K-Means clustering to the data points. 3. Clusters are plotted in different colors for readability.
10.	Alternative Path: Pick different clustering algorithms.	
11.	Exception: Data is too similar so the clusters are not well defined.	

Table 9 Clustering Narrative

1.	Use Case Name: Select Audio File	
2.	Implementation Priority: 7	
3.	Actors: User	
4.	Summary: Pick the audio that needs to be classified.	
5.	Pre-condition: Audio should be clear and in proper format.	
6.	Post-condition: User can view reports on the data.	
7.	Extend: Audio file can be uploaded.	
8.	Include: -	
9.	Normal Course of Events:	
	System Actor Action	System Response
	1. Admin picks a file from the device.	2. System runs all the backend code and will give the user the final result.
10.	Alternative Path:-	
11.	Exception: Wrong format files being input.	

Table 70 Select Audio File Narrative

3.3 Functional Requirements

- Urdu Speech Corpus Generation
- Speech to Text conversion
- Data Pre-Processing
    - Tokenization
    - Removal of white spaces and stop words
    - Lemmatization
    - Lexicon
    - Word & Sentence Embedding
- Sentiment Analysis
- Clustering
- Display result through front end

3.4 Non-Functional Requirements

Performance Requirements:
The system should run smoothly without lagging in any module. Should be able to handle every scenario and deal with extreme values.

Security Requirements
The data being used for classification should remain private and anonymous to anyone running the program unless the data is flagged as dangerous.

Software Quality Attributes

- **Flexibility**: - In context of user, it is most important to know what kind of flexibility the user has with the system that they are using. However the user wants to enter his data, the option should suit him needs.
- **Maintainability**: - The end user should also be provided with some kind of help tool so the system can be maintained.
- **Usability**: - It is important to have reliable and repeatable evaluation techniques. Without them we can't tell developers if their system is responding well or not.
- **Correctness**: -The system should be correct with the analysis it displays to the user. There should be no room for error.

CHAPTER 4

4. Methodology

4.1 Speech Corpus

The audio clips are collected in .wav format of length 10 seconds to maintain a set standard. Since we could not get pre-recorded calls, we improvised and recorded live conversions from various people through a microphone which assists us in canceling noise in recordings. Furthermore, some of the recordings are scripted to meet the requirement of our testing system to ensure maximum accuracy. Online resources are used to populate the dataset as well. The clips are at a SAMPLE RATE 44.1 kHz, 16-bit PCM amplitude normalized. The general public is approached for these sound bites to ensure that there was diversity in the speaking patterns being recorded. Once collected, the files are run through the Audacity software to remove any excess noise and dips in audio.

Data	Files	Duration	Average Bit-Rate
18 hours	7000	10 seconds	1378.12

Table 81 Audio Meta Data

4.2 Pre-Processing

The Google Speech Recognition API is used to convert the audio files into text. To apply conditions and search words in a sentence, the files were converted into Urdu. The alphabet is written right-to-left for the Urdu language. Urdu is a modification of the Persian alphabet known as Perso-Arabic, which is itself a derivative of the Arabic alphabet. The Urdu alphabet has up to 58 letters with 39 basic letters and no distinct letter cases, the Urdu alphabet is typically written in the calligraphic Nastaʿlīq script. There are 46 Alphabets, 10 Digits, 6 Punctuations, and 6 Diacritics. Once the text files are stored, we will apply NLP procedures to normalize the data and remove unnecessary information.

The initial step is to tokenize the data, the result of which is a list of all the words present in the audio clip in the shape of tokens. The next step is the cleaning of the data which aims to clean input text from punctuation marks, HTML stripping, URLs, and other special characters to prepare the text for further processing in the sentiment analysis module. In this step, the text is also normalized to ensure the uni-codes being used are of the same language and are maintaining uniformity. Due to the orthographic characteristics of Urdu text, such as the optional use of diacritics and the ambiguity in word boundaries, two additional tasks namely: diacritic omission, and word boundary identification, are added in the text cleaning process. Diacritics are optional in Urdu, depending on the author. For example, (ا، آ، أ), is a regular practice to remove them during text normalization. When there are multiple characters from different languages which look the same and sound but different uni-codes. Now as computers deal with numbers, the same character appearing in more than one place in a different language will have a different Unicode and that will create confusion which will create problems in understanding the context of the data. ['ف', 'ﻗ', 'ﻘ'] are all converted to ‘ف’.

The next step is to remove white spaces and the stop words. For this project, we created and gathered a dataset of Urdu stop words based on which we removed words. Followed by lemmatization to reduce the complexity of the sentence and shrink words down to their base. Lemmatization is the process where we take individual tokens from a sentence and we try to reduce them to their base form. The output of the lemmatization process is the base form of the word (lemma). The root/stem of the word, a lemma is used to get the vocabulary form of a word which helps in understanding the roots of different words. Lemmatization refers to doing things properly with the use of a vocabulary and morphological analysis of words, normally aiming to remove inflectional endings only and to return the base or dictionary form of a word, which is known as the lemma. This helps normalize text for text classification tasks or search engines, and a variety of other NLP tasks such as sentiment classification, which we will be covering in this paper.

The pre-processing module is concluded by sentence embedding. As the data is unsupervised, we need to apply clustering in the later stages to be able to classify data. Using a pre-trained Word2Vec model, we convert all our words into 300-dimensional vectors, which are then averaged to calculate the sentence vectors

Sentence	Key Words	Stop Words
ایک نئی تحقیق میں کہا گیا ہے کہ دوسری جنگ عظیم میں اتحادی افواج کی جانب سے جو بم برسائے گئے وہ اتنے طاقتور تھے کہ ان سے کرۂ ہوائی کو نقصان پہنچا ہے۔	تحقیق، دوسری جنگ عظیم، اتحادی افواج، بم، برسائے، طاقتور، کرۂ ہوائی، نقصان، پہنچا	ایک ، نئی، میں، کہا، گیا، ہے، کہ، کی، جانب، سے، جو، گئے، وہ، اتنے، تھے، ان، کو،
برطانیہ میں قائم لیورپول یونیورسٹی کے پروفیسر فرانسس مکلیون نے بتایا کہ مچھر یا دیگر کیڑے ' مکوڑے اور پودے انسانی جلد پر ٹاکسن یعنی حیوانی یا نباتاتی زہر چھوڑتے ہیں۔ جسکی وجہ سے اعصاب کی جانب سے دماغ کو خارش کا سگنل ملتا ہے اور ہم خارش کرنے لگتے ہیں۔	برطانیہ، قائم، لیورپول یونیورسٹی، پروفیسر، فرانسس مکلیون، مچھر، کیڑے مکوڑے، پودے، انسانی، جلد، ٹاکسن، حیوانی، نباتاتی، زہر، چھوڑتے، اعصاب، دماغ، خارش، سگنل	میں، کے، نے، بتایا، کہ، یا، دیگر، اور، پر، یعنی، ہیں، جسکی، وجہ، سے، کی، جانب، کو، کا، ملتا، ہے، ہم، کرنے، لگتے

Table 12 Example of Urdu Stop Words and Key Words

4.3 Sentiment Analysis

It is the process of identifying, extracting, and quantifying the meaning behind a sentence. Two main approaches are used for sentiment analysis i.e. machine learning approach and the lexicon-based approach. In the machine learning approach, the data is classified by training a classifier on the labeled data. In the lexicon-based approach, lexical items from dictionaries are assigned positive and negative polarities. The lexicon is composed of a defined list of words with their corresponding sentiments and can include their intensities and polarities as well. The lexicon-based approach is considered simple and reliable as compared to the machine learning approach since it avoids the need to develop a labeled training set. Moreover, it is also difficult to ensure the correctness of labeled data in the machine learning approach. To develop a sentiment lexicon, some researchers use a dictionary or corpus-based approach. Corpus-based approaches involve determining the patterns of co-occurring of words to determine the sentiments of the words and phrases. The dictionary-based approach helps to compile the sentiment words and use the antonyms and synonyms in the file to determine the sentiments of the lexical items. Depending on the nature of the data and the choice of the users, the process of sentiment analysis can be performed on three different levels. These three levels of analysis are;

1) document-level sentiment analysis,

2) aspect-level sentiment analysis and

3) sentence-level sentiment analysis.

This paper is done at sentence level sentiment analysis where sentences are classified as positive, negative, or neutral. Here the sentence is considered as a separate unit expressing a single opinion.

The following algorithm was used for automatically tagging the sentiment of a given input sentence

a) Remove special symbols and punctuations from the sentence

b) Compute total positive words (pw) in a sentence

c) Compute total negative words (nw) in a sentence

d) Now assign sentiment to X using the following rules

i. If pw > nw, assign positive sentiment to X

ii. If nw > pw, assign negative sentiment to X

iii. if pw = pw, assign neutral sentiment to X after the application of the above algorithm, the accuracy of the lexicon was determined.

4.4 Clustering

The data used for this project is unlabeled due to the fact that we are generating it ourselves, the process of sentiment analysis will not be able to deduce context from the data available. For that reason, the method of K-Mean Clustering was adopted to be able to classify our data based on its context.

The concept is to classify the given set of data into a selected number of clusters. The algorithm consists of two separate phases: the first phase is to define the centroids, one for each cluster. The next phase is to take each point belonging to the given data set and associate it to the nearest centroid. When all the points are defined to some clusters, that is the initial step completed and the early grouping is done. At this point, we need to recalculate the new centroids, as the inclusion of new points may lead to a change in the cluster centroids.

What determines a "good," or representative, clustering? Consider a single cluster of points along with its centroids or mean. If the data points are tightly clustered around the centroids, the centroids will be representative of all the points in that cluster. The standard measure of the spread of a group of points about its mean is the variance or the sum of the squares of the distance between each point and the mean. If the data points are close to the mean, the variance will be small. A generalization of the variance, in which the centroid is replaced by a reference point that could be a centroid, is used in cluster analysis to indicate the overall quality of the partitioning; specifically, the error measure E is the sum of all the variances:

$$E = \sum_{i=1}^{k} \sum_{j=1}^{n_i} \|x_{i,j} - z_i\|^2$$

4.5 Process Flow Representation

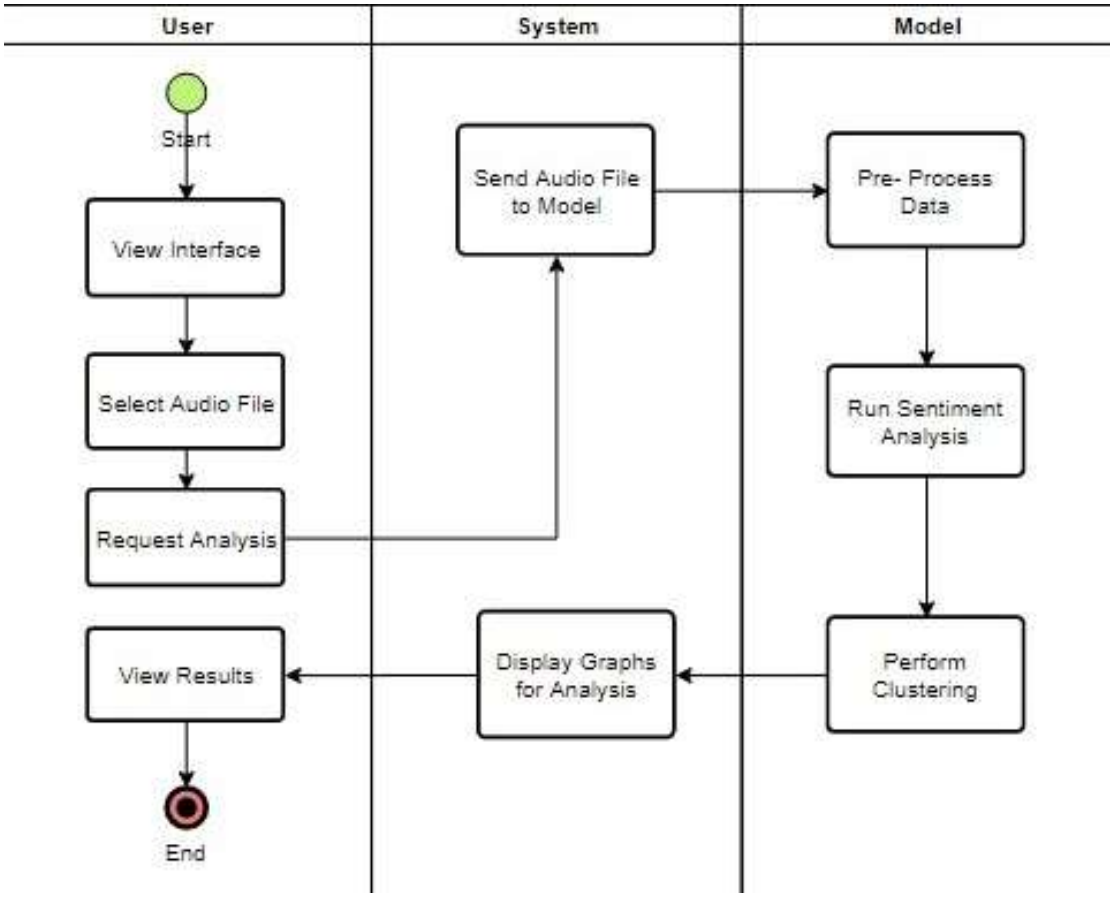

Figure 3 Swim Lane

4.6 Process Sequence Representation

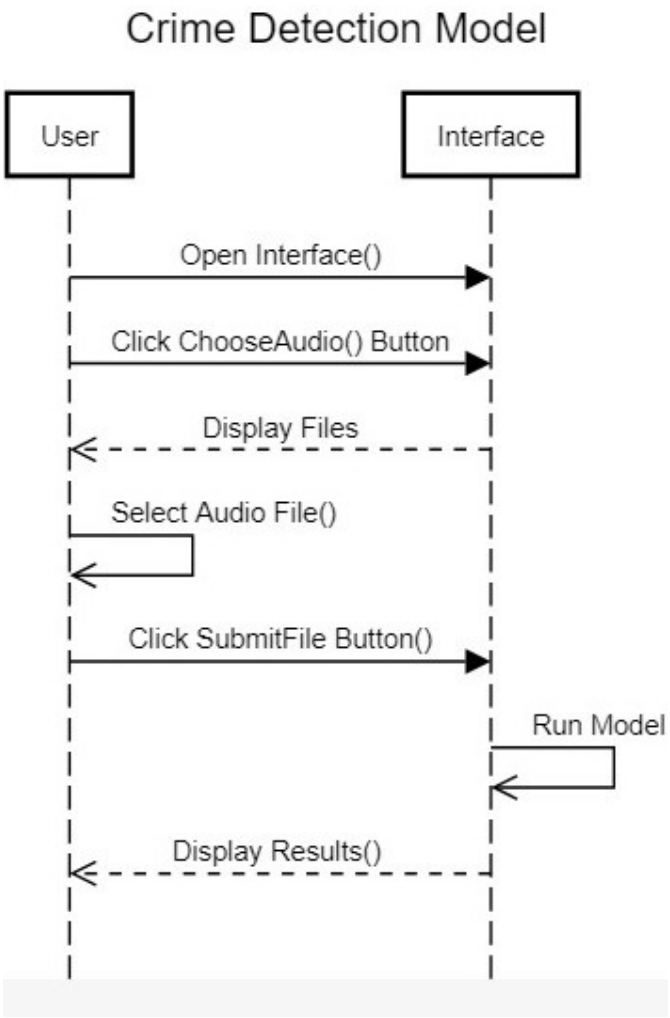

Figure 4 Sequence Diagram

4.7 Work Flow Diagram

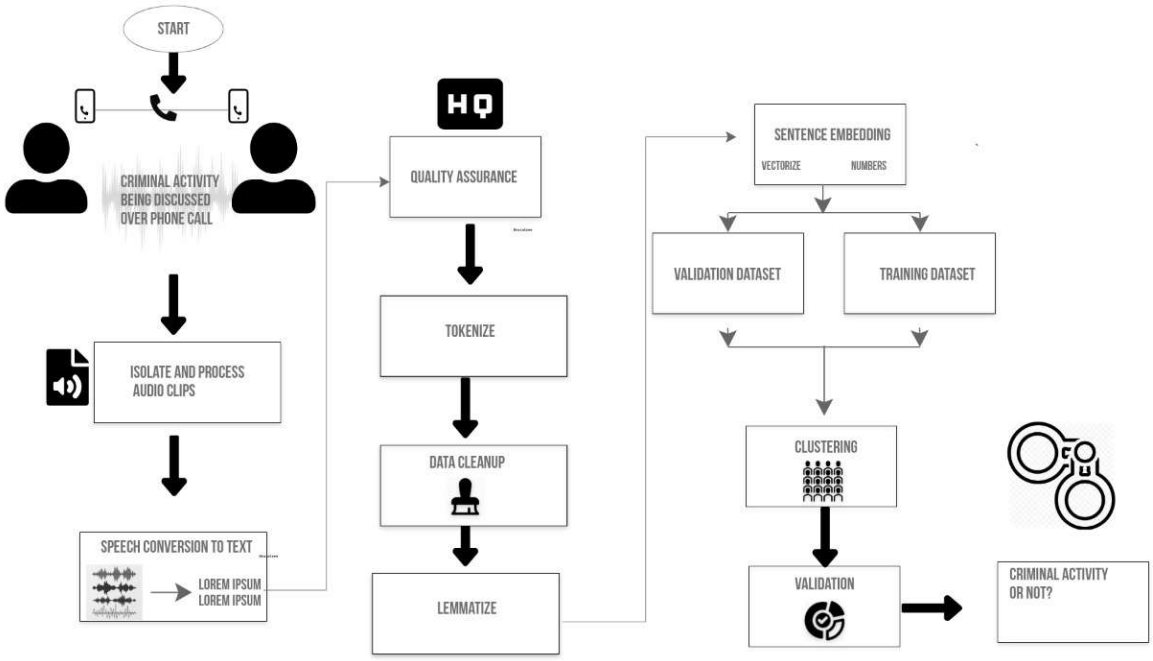

Figure 5 Work Flow Diagram

CHAPTER 5

5 Experimental Setup

This chapter describes all the practical aspects of the works done in the project. The audio dataset used in this project was a mixture of self-generated files and also online resources. We collect over 20 hours of audio clips. The clips are at a SAMPLE RATE 44.1 kHz, 16-bit PCM amplitude normalized. The audio clips were edited using Audacity software.

In all the following operations, the audio files are first converted to text and after that, the data pre-processing is applied. Speech to text conversion is done using the Google Speech Recognition API. Pre-processing is done with the help of lexicons which are self-generated as well as gained from online resources. The data is cleaned and sentiment is gathered on basis of lexicons. Once we have sentiment, the sentences are embedded using a pre-trained Word2Vec model. It converts each word into 300 dimensional vectors, which we then use to calculate sentence average vectors. These vectors are then plotted on a graph using plt.scatter() and then classified using the K-Means clustering algorithm. We find 'k' new centroids, a new grouping is to be created between the same data points and the nearest new centroid, generating a loop. As a result of this loop, the k centroids can change their position step-by-step. Eventually, a situation will be reached where the centroids do not move anymore. This signifies the convergence criterion for clustering.

The k-means algorithm is the most extensively studied clustering algorithm and is generally effective in producing good results. The major drawback of this algorithm is that it can produce different clusters for a different set of values for the initial centroids. The quality of the final clusters heavily depends on the selection of the initial centroids. To determine the optimal value of k clusters, we use the elbow method. The elbow method is expressed by Sum of Squared Error. SSE is the sum of the squared differences between each observation and its group's mean. It can be used as a measure of variation within a cluster. If all cases within a cluster are identical the SSE would then be equal to 0. The formula is written as:

$$SSE = \sum_{i=1}^{n} (x_i - \bar{x})^2$$

Where n is the number of observations xi is the value of the ith observation and x bar is the mean of all the observations.

To validate the clusters that form and their accuracy, internal validation methods such as the Silhouette Coefficient were applied. Silhouette refers to a method of interpretation and validation of consistency within clusters of data. The technique provides a succinct graphical representation of how well each object has been classified. The sil-

houette value is a measure of how similar an object is to its cluster compared to other clusters.

Silhouette Coefficient or silhouette score is a metric used to calculate the goodness of a clustering technique. Its value ranges from -1 to 1.

1: Means clusters are well apart from each other and distinguished.
0: Means clusters are indifferent, or we can say that the distance between clusters is not significant.
-1: Means clusters are assigned in the wrong way.

$$\text{Silhouette Score} = \frac{(b-a)}{\max(a \cdot b)}$$

where

a= average intra-cluster distance i.e the average distance between each point within a cluster.
b= average inter-cluster distance i.e the average distance between all clusters.

That is, we want the average distance within a cluster to be as small as possible; the average distance between clusters to be as large as possible.

Apart from data pre-processing and clustering, an interface is also required, where users can upload his/her audio clip and after processing, results can be displayed. Hence; we have used Streamlit to develop a web interface. Streamlit is an open-source Python library that makes it easy to create and share custom web applications for machine learning and data scientists. In just a few minutes, powerful data applications can be deployed and build.

Select Audio File:

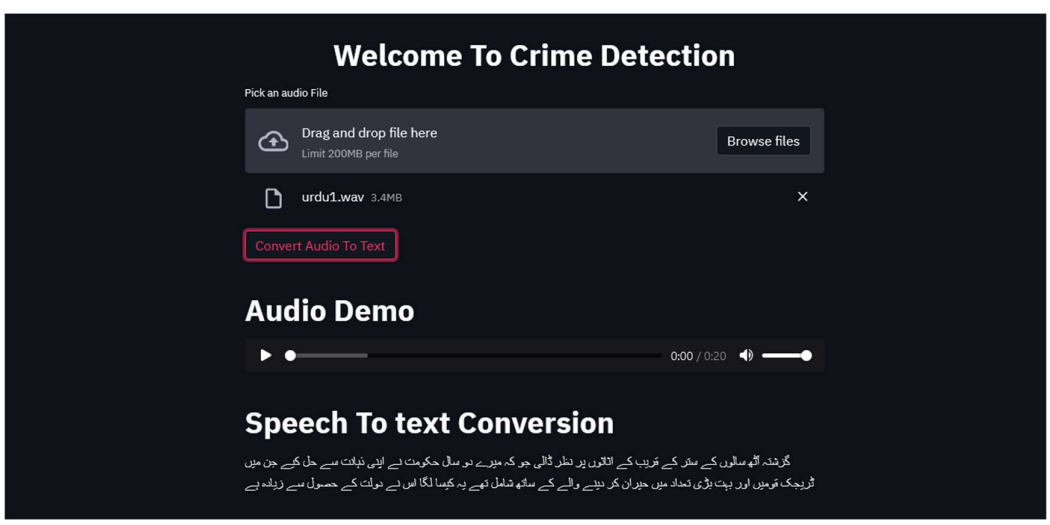

Figure 6 Select Audio File

Result of Pre-Processing:

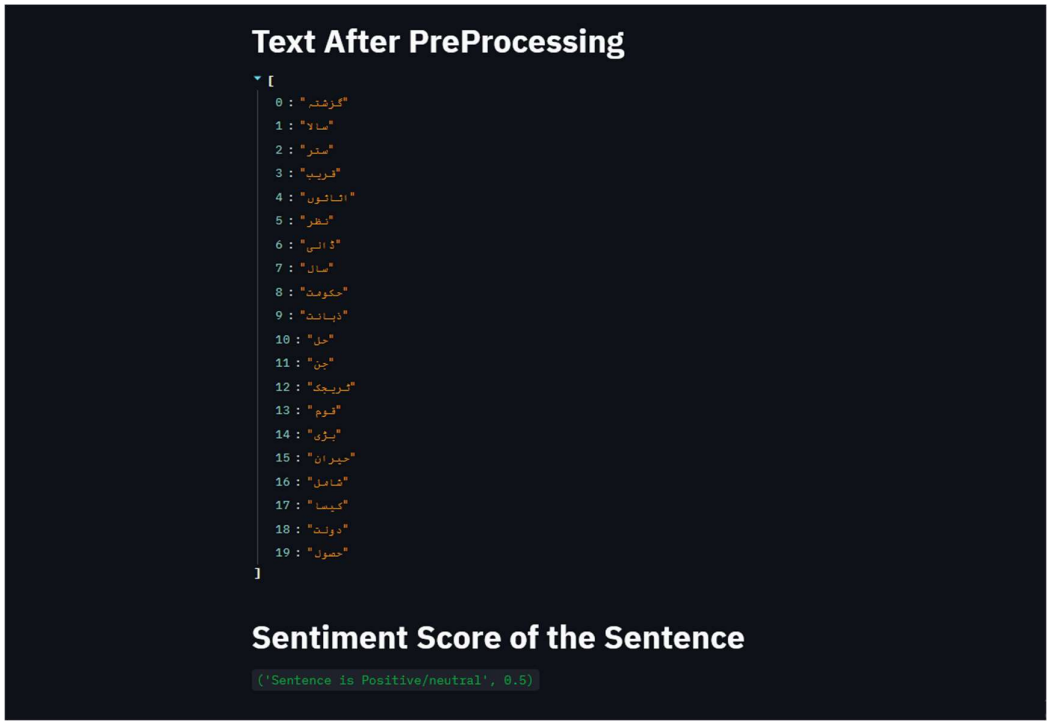

Figure 7 Pre-Processing Results

The only system requirements are Python 3.6 - Python 3.8. As its web-based, so it can be accessed through a browser link via laptop or a PC as well as through a smartphone. The libraries that we used were Python 3.7. Python is chosen over Matlab because it is easier to install and readily available. Python has extensive free libraries and API's that support NLP processes and machine learning, which makes it an ideal language for the project. Moreover, the open-source library for Python provides various implementations of data processing algorithms used extensively in the project.

CHAPTER 6

6. Results and Conclusions

6.1 Results

Working on the audio files, it was quickly established that the conversion to text was only going to maintain accuracy if the length of the clips was short. Ten seconds of audio resulted in an accuracy of roughly 89%. Ten seconds could be used to run one sentence through the system. The accuracy was calculated by manually checking sentences where the audio files were compared to the translated text and the differences were marked. It was important to ensure that the pronunciation of words was clear and also the pauses were not too long as the system would stop reading file.

Once textual data was obtained, the pre-processing module cleaned and reduced the size of the sentences. Clustering was done on the plotted data, the average sentiment score of each cluster provided us with an overview of our dataset in terms of the polarity of their contexts. Sentences were separated into clusters based on their vector scores where they were further verified by lexicon based sentiment analysis and each cluster is assigned an average sentiment score as shown in Table. 13.

Cluster #	Average Sentiment Score	Sample
Cluster 1	0.8	علی ایک انتہائی بے لوث شخص ہے جو ہمیشہ ضرورت مندوں کی مدد کرتا ہے۔
Cluster 2	-0.2	- میں نے اسے نیند میں قتل کیا۔ - میں نے اسے لوٹا کیونکہ اس کے پاس بہت پیسہ ہے۔
Cluster 3	0.04	آج موسم ٹھیک تھا ، تھوڑی سی ہوا کے ساتھ
Cluster 4	0.4	یہ ایک دلچسپ کتاب ہے جس میں پاکستان کی تاریخ پر بحث کی گئی ہے۔

Table 13 Results of Clustering

6.2 Conclusion

Urdu is a very low resource language so the purpose of this research was to identify an effective way to deal with unlabeled audio dataset in Urdu. Based on the analysis done, it can be concluded that the peculiarities of Urdu such as lack of resources, rich morphology, lack of capitalization and many other tasks makes Urdu document clustering more complex when compared with the language having script writing style from left to right. Due to the unavailability of labeled data and the time constraints, the level of sentiment classification was not definitive or absolute, which can only be achieved for this method with complete positive and negative lexicons which cover the entire Urdu vocabulary.

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